

PEAS Framework and Task Environment Analysis

AI Tutor for Personalized Student Learning Recommendations

Task 1: PEAS Specification

Performance Measures

The success of the AI Tutor system can be measured using the following metrics:

Recommendation Accuracy — How accurately the system recommends learning materials and topics.

Student Improvement Rate — Improvement in student scores after following recommendations.

User Satisfaction Score — Feedback rating provided by students.

Response Time — Time required to generate recommendations.

Learning Goal Achievement Rate — Percentage of students achieving their learning objectives.

Environment

The AI Tutor operates in educational environments such as schools, colleges, universities, and online learning platforms. The environment includes students with different learning abilities, subjects, learning styles, academic histories, and performance levels. External factors include internet connectivity, curriculum changes, and varying student engagement levels.

Actuators

The AI Tutor interacts with users through:

Personalized Study Plan Generator

Learning Resource Recommendation Engine

Progress Notifications and Alerts

Performance Feedback System

Quiz and Practice Question Recommendations

Sensors

The AI Tutor gathers information using:

Student Quiz Scores

Assignment Results

Learning History Records

Subject Preferences

Time Spent on Learning Activities

Student Feedback and Ratings

Task 2: Environment Classification

1. Fully Observable vs. Partially Observable

Choice: Partially Observable

Justification: The tutor cannot completely observe a student's motivation, concentration level, or future behavior, making some important information unavailable.

2. Single-Agent vs. Multi-Agent

Choice: Single-Agent

Justification: The AI Tutor is the primary decision-making entity responsible for generating recommendations for students.

3. Deterministic vs. Stochastic

Choice: Stochastic

Justification: Student performance and learning outcomes are uncertain and can vary due to many unpredictable factors.

4. Episodic vs. Sequential

Choice: Sequential

Justification: Current recommendations depend on previous learning activities and influence future learning outcomes.

5. Static vs. Dynamic

Choice: Dynamic

Justification: Student knowledge and performance continuously change over time, requiring updated recommendations.

6. Discrete vs. Continuous

Choice: Continuous

Justification: Variables such as scores, study time, engagement levels, and performance metrics can take a wide range of values.

Task 3: Critical Analysis

Question 1

Which environment dimension poses the greatest technical challenge?

The greatest technical challenge for an AI engineer is the Partially Observable nature of the environment. An AI Tutor can only access measurable information such as scores, attendance, and activity logs. Important factors such as student motivation, emotional state, concentration, personal circumstances, and future learning behavior cannot be directly observed. As a result, the system must make recommendations using incomplete information, which can reduce prediction accuracy and recommendation quality.

Question 2

Proposed Utility Function

$$U = 0.7(\text{Student Improvement}) + 0.3(\text{Student Engagement}) - 0.2(\text{Time Cost})$$

This utility function balances learning effectiveness and student engagement while minimizing excessive study time.

If the weight of Student Improvement is doubled, the system will prioritize academic performance much more aggressively. The tutor may recommend additional exercises, more challenging tasks, and longer study plans. While this could improve learning outcomes, it may also reduce student engagement due to increased workload.

Task 4: Structured Representation (JSON)

```
{
  "system_name": "AI Tutor for Personalized Student Learning Recommendations",
  "peas": {
    "performance_measures": [
      "recommendation_accuracy",
      "student_improvement_rate",
      "user_satisfaction_score",
      "response_time",
      "learning_goal_achievement_rate"
    ],
  },
}
```

```

"environment": "Schools, universities, and online learning platforms with students of
varying abilities and learning styles",
"actuators": [
    "study_plan_generator",
    "learning_resource_recommender",
    "progress_notifications",
    "performance_feedback_system",
    "quiz_recommendation_engine"
],
"sensors": [
    "quiz_scores",
    "assignment_results",
    "learning_history",
    "subject_preferences",
    "study_time_records",
    "student_feedback"
]
},
"environment_classification": {
    "observability": {
        "choice": "Partially Observable",
        "justification": "Student motivation and future behavior cannot be fully observed."
    },
    "agents": {
        "choice": "Single-Agent",
        "justification": "The AI Tutor is the primary decision-making entity."
    },
    "determinism": {
        "choice": "Stochastic",
        "justification": "Learning outcomes are influenced by uncertain factors."
    },
    "episodic_sequential": {
        "choice": "Sequential",
        "justification": "Current recommendations affect future learning outcomes."
    },
    "static_dynamic": {
        "choice": "Dynamic",
        "justification": "Student performance changes continuously over time."
    },
    "discrete_continuous": {
        "choice": "Continuous",
        "justification": "Scores, study time, and engagement values vary continuously."
    }
},
"utility_function": "U = 0.7(Student Improvement) + 0.3(Student Engagement) - 0.2(Time Cost)"
}

```

Conclusion

The AI Tutor is an intelligent educational system that analyzes student performance and learning behavior to generate personalized recommendations. By adapting to individual needs, the system can improve learning outcomes, increase engagement, and support more effective education.